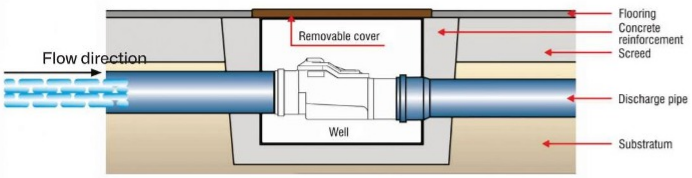
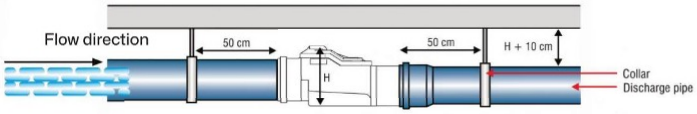
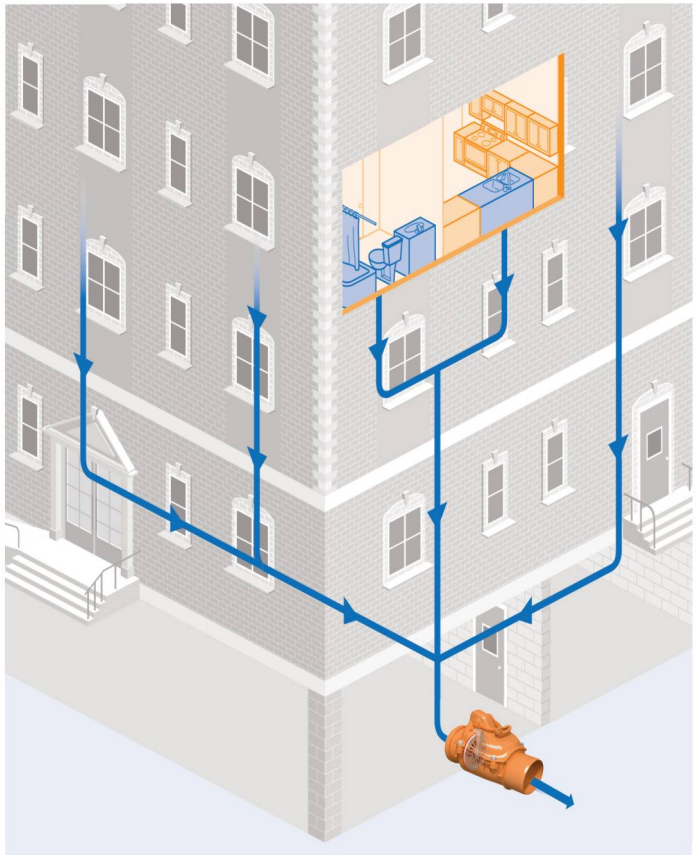
Installing the non-return valve

The non-return valve must be positioned upstream of the main manifold in a suitable accessible well



or by intercepting a pipe suspended horizontally on the intrados of the slab





About Dadex

Dadex was established as a public limited company in 1959. We started with the manufacturing of FC roofing systems at our facility in Hyderabad, under the dynamic leadership of our Founder Chairman, Mr. Kassim Dada. Being a forward-looking organization, his son, Mr. Sikander Dada kept up to date with the market, and was instrumental in the diversification with plastic piping systems. Presently we produce various thermoplastic systems such as UPVC, PP-R, PP-RCT, PP, PE.

Headquartered in Karachi, our two manufacturing facilities based in Hyderabad and Sundar Industrial Estate (near Lahore) are equipped with state of the art machinery. Our joint efforts have led Dadex to the ISO 9001, ISO 14001, ISO 45001 and PSQCA certifications.

We aim to set standards in every field we enter. Customer focus, quality and innovation are reflected in every aspect of our business. Offering customized solutions to a cross section of markets for over 66 years, Dadex today is a hallmark of trust and reliability.

We believe that our strength lies in delighting our customers. That is why providing quality products backed by round the clock technical support services are the Dadex promise to all our valued customers.

DADEX Eternit Limited

Dadex House, 34-A/1, P.E.C.H.S. Block-6, Shahrah-e-Faisal, Karachi. UAN: (92-21) 111 000 789,
Email: marketing@dadex.com.pk

For more information, please visit www.dadex.com. For Technical information and assistance, please contact Dadex Techno Commercial Department at sales@dadex.com.pk

Feb. 2026

FlowLine

SEWERAGE AND DRAINAGE

NON-RETURN VALVE



UPVC



SPECIALLY DESIGNED TO ELIMINATE THE RISK OF FLOOD DAMAGE CAUSED BY THE BACKFLOW OF SEWAGE OR RAIN WATER

magnaplast

DADEX

Introduction

We could blame it on climate change or on the "irresponsibility" of men, but it is a fact that more and more often, the rains that hit our country cause considerable problems of flooding in buildings. The sudden and massive fall of several millimeters of water does not find a correct outlet and this generates water returns inside the buildings.

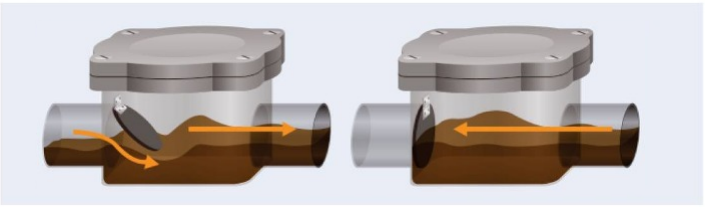
Very often, despite the knowledge of the reasons that cause these problems, it is not possible to radically intervene on the sewer network and, therefore, it becomes necessary to adopt a solution directly on the building to be protected from flooding risks. This solution is called Non-return valve.

What is it and what are the causes of reflux

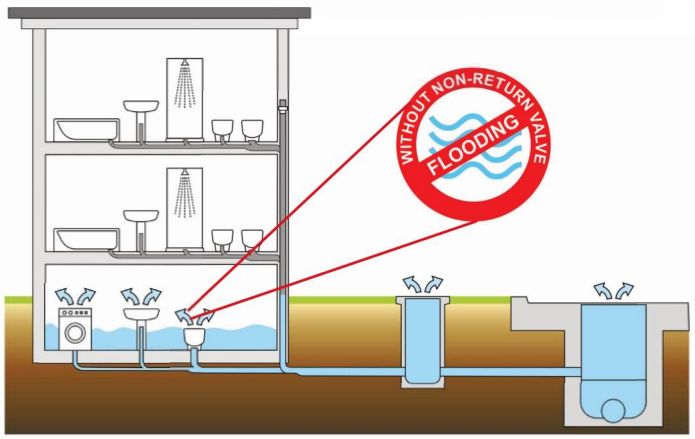
Knowing the principle of "communicating vessels", according to which water tends to maintain the same level in all branches of a hydraulic circuit, when we talk about reflux, we define the maximum water level of a hydraulic circuit beyond which the leaking from the circuit itself. For this reason, if the sanitary utilities are below the ebb level (basically basements, cellars, garages, but also living quarters that are in any case below the level of the road level), flooding can occur during overload of the sewer network.

The reasons for the reflux effect may be the following:

- An incorrectly designed drainage system.
- High hydraulic flows caused by short but very intense rainfall.
- Increase in the volumes of rain channeled by collectors in relation to the building increase and urbanization.
- Water overload due to contemporaneity peaks in areas with high urban density.
- Malfunctions and occlusions downstream of the drainage system.



Backflow without the presence of NRV

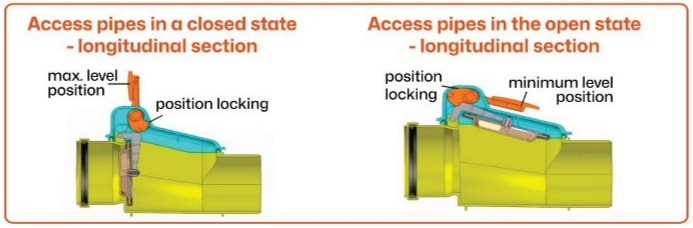


The non-return valve as a protection and safety solution for homes

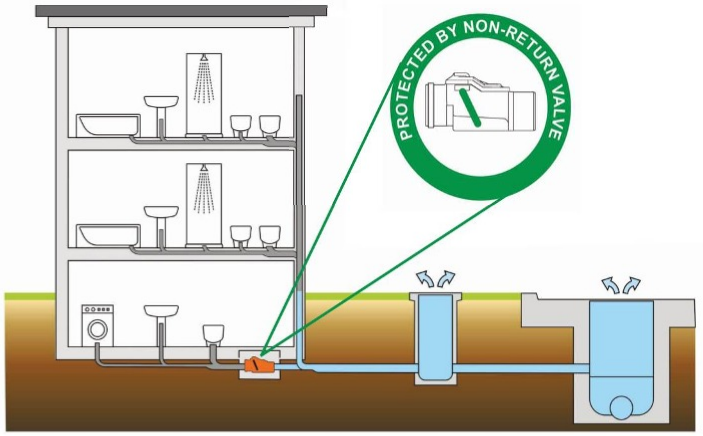
The non-return valve is designed for use within sewerage & drainage systems to eliminate the risk of flood damage caused by the backflow of sewage or flood water into properties through low-level entries, such as drain gullies and toilets. It is a reliable solution for protecting against the backflow of sewage and rainwater.

This valve is designed for effective water flow control, ensuring that water only flows in one direction, thereby minimizing the risk of flooding during heavy rainfall.

The non-return valve is housed in a suitable accessible and inspectable well. Its role is to create an effective barrier to any return of water flow coming from the sewerage system during any possible overload event. The non-return valve is equipped with a stainless steel plate that pivots in the direction of the water flow from the building to the sewer system. In the case of an ebb, the plate closes mechanically with the thrust of the water. The plate, acting on a special lever, can be permanently hermetically locked.



Backflow with the presence of NRV



Furthermore, its use is absolutely recommended in existing buildings, where it is possible to intervene on the connection piping to the sewer network and there is the physical space to create a pit that houses the valve. The non-return valve must be approved in accordance with the European standard EN 13564-1: 2003 which guarantees its requirements, such as: effective flow, non-deformability, resistance, hydraulic seal, compliance of connection, correct opening /closing operation.

Main Features

- High wall smoothness.
- Very low weight.
- Simple and quick assembly.
- Possibility and ease of combining with other materials.
- Total resistance to general and pitting corrosion.
- High durability, estimated for 100 years.
- Resistance to harmful effects of chemical compounds
- Resistance to deposits on the inner surfaces of pipes

Material: Unplasticised Polyvinyl Chloride (UPVC)

Standard: EN 1401-1

Available Sizes: 110mm, 160mm and 200mm